3 MATRITSA RANGI. TESKARI MATRITSA

3.1. $A=\begin{pmatrix}2 & 3 & 2\\ 5 & 1 & 4\\ 1 & -2 & -1\end{pmatrix}$ matritsa uchun teskari A^{-1} matritsani klassik usulda toping.

Yechish. Klassik usulda teskari matritsa

$$A^{-1}=\frac{1}{|A|}\begin{pmatrix}A_{11} & A_{21} & A_{31}\\ A_{12} & A_{22} & A_{32}\\ A_{13} & A_{23} & A_{33}\end{pmatrix} \qquad (1)$$

formula bo‘yicha hisoblanadi.

Bu yerda $|A|$ berilgan matritsa determinanti,

$A_{ij}\ (i=1,2,3;\ j=1,2,3)$ A matritsa elementlarining algebraik to‘ldiruvchilari.

$$|A|=\begin{vmatrix}2 & 3 & 2\\ 5 & 1 & 4\\ 1 & -2 & -1\end{vmatrix}=-2+12-20-2+15+16=43-24=19\neq 0$$

Demak A xosmas matritsa, va A^{-1} teskari matritsa mavjud.

Algebraik to‘ldiruvchilarni hisoblaymiz:

$$A_{11}=\begin{vmatrix}1 & 4\\ -2 & -1\end{vmatrix}=-1+8=7;\quad A_{12}=-\begin{vmatrix}5 & 4\\ 1 & -1\end{vmatrix}=-(-5-4)=9;$$

$$A_{13}=\begin{vmatrix}5 & 1\\ 1 & -2\end{vmatrix}=-10-1=-11;\quad A_{21}=-\begin{vmatrix}3 & 2\\ -2 & -1\end{vmatrix}=-(-3+4)=-1;$$

$$A_{22}=\begin{vmatrix}2 & 2\\ 1 & -1\end{vmatrix}=-2-2=-4;\quad A_{23}=-\begin{vmatrix}2 & 3\\ 1 & -2\end{vmatrix}=-(-4-3)=7;$$

$$A_{31}=\begin{vmatrix}3 & 2\\ 1 & 4\end{vmatrix}=12-2=10;\quad A_{32}=-\begin{vmatrix}2 & 2\\ 5 & 4\end{vmatrix}=-(8-10)=2;$$

$$A_{33}=\begin{vmatrix}2 & 3\\ 5 & 1\end{vmatrix}=2-15=-13;$$

A_{ij} larni (1) formulaga qo‘yamiz va teskari

$$A^{-1}=1/19\begin{pmatrix}7 & -1 & 10\\ 9 & -4 & 2\\ -11 & 7 & -13\end{pmatrix}$$ matritsani olamiz.

Teskari matritsaning to’griligini tekshirish uchun quyidagi tenglikni tekshiramiz:

$$AA^{-1}=A^{-1}A=E \qquad (2)$$

$$\begin{pmatrix}2 & 3 & 2\\ 5 & 1 & 4\\ 1 & -2 & -1\end{pmatrix}\cdot 1/19\begin{pmatrix}7 & -1 & 10\\ 9 & -4 & 2\\ -11 & 7 & -13\end{pmatrix}=$$

$$=1/19\cdot\begin{pmatrix}14+27-22 & -2-12+14 & 20+6-26\\ 35+9-44 & -5-4+28 & 50+2-52\\ 7-18+11 & -1+8-7 & 10-4+13\end{pmatrix}=$$

$$=1/19\cdot\begin{pmatrix}19 & 0 & 0\\ 0 & 19 & 0\\ 0 & 0 & 19\end{pmatrix}=\begin{pmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}=F$$

Demak, A^{-1} to‘g‘ri topilgan. ▶

3.2. $A=\begin{pmatrix}1 & 2 & 1\\ -1 & -1 & -3\\ 4 & 3 & -2\end{pmatrix}$ matritsa uchun A^{-1} matritsani Gauss-Jordan usulida toping.

Yechish. $|A|=16\neq 0$ teskari matritsa mavjud. Berilgan matritsani birlik matritsa hisobida kengaytirib, elementar almashtirishlar bajaramiz, bu usulni to chap tomonda A matritsa o‘rnida birlik matritsa hosil bo‘lguncha davom ettiramiz, o‘ng tomonda hosil bo‘lgan matritsa berilgan matritsaga nisbatan teskari matritsa bo‘ladi.

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ -1 & -1 & -3 & 0 & 1 & 0 \\ 4 & 3 & -2 & 0 & 0 & 1 \end{array}\right) \sim \left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & -2 & 1 & 1 & 0 \\ 0 & -5 & -6 & -4 & 0 & 1 \end{array}\right) \sim$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & -2 & 1 & 1 & 0 \\ 0 & 0 & -16 & 1 & 5 & 1 \end{array}\right) \sim \left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & -2 & 1 & 1 & 0 \\ 0 & 0 & 1 & -1/16 & -5/16 & -1/16 \end{array}\right) \sim$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & 0 & 5 & -1 & -2 & 0 \\ 0 & 1 & 0 & 14/16 & 6/16 & -2/16 \\ 0 & 0 & 1 & -1/16 & -5/16 & -1/16 \end{array}\right) \sim \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -11/16 & -7/16 & 5/16 \\ 0 & 1 & 0 & 14/16 & 6/16 & -2/16 \\ 0 & 0 & 1 & -1/16 & -5/16 & -1/16 \end{array}\right)$$

$A^{-1} = 1/16 \begin{pmatrix} -11 & -7 & 5 \\ 14 & 6 & -2 \\ -1 & -5 & -1 \end{pmatrix}$ teskari matritsa to‘g‘ri topilganini (2) formulaga qo‘yib

tekshiramiz: $AA^{-1} = 1/16 \begin{pmatrix} 1 & 2 & 1 \\ -1 & -1 & -3 \\ 4 & 3 & -2 \end{pmatrix} \cdot \begin{pmatrix} -11 & -7 & -5 \\ 14 & 6 & -2 \\ -1 & -5 & -1 \end{pmatrix} =$

$$= 1/16 \begin{pmatrix} -11+28-1 & -7+12-5 & 5-4-1 \\ 11-14+3 & 7-6+15 & -5+2+3 \\ -44+42+2 & -28+18+10 & 20-6+2 \end{pmatrix} =$$

$= 1/16 \begin{pmatrix} 16 & 0 & 0 \\ 0 & 16 & 0 \\ 0 & 0 & 16 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ demak, teskari matritsa to‘gri topilgan.

▶

? Teskari matritsani toping.

3.3. Berilgan kvadrat matritsalar uchun teskari matritsani ikki usulda toping:

$$a)\begin{pmatrix} -1 & 1 \\ 4 & 2 \end{pmatrix}; \quad b)\begin{pmatrix} 1 & 3 \\ 2 & 6 \end{pmatrix}; \quad c)\begin{pmatrix} tg\alpha & 1 \\ 2 & ctg\alpha \end{pmatrix}; \quad d)\begin{pmatrix} 2 & 1 & 1 \\ 1 & 0 & 2 \\ 3 & 1 & 2 \end{pmatrix}$$

$$e)\begin{pmatrix} 1 & -1 & 1 \\ -38 & 41 & -34 \\ 27 & -29 & 24 \end{pmatrix};\quad f)\begin{pmatrix} 2 & 5 & 7 \\ 6 & 3 & 4 \\ 5 & -2 & -3 \end{pmatrix};\quad j)\begin{pmatrix} 3 & -4 & 5 \\ 2 & -3 & 1 \\ 3 & -5 & -1 \end{pmatrix}$$

3.4. Berilgan kvadrat matritsalar uchun teskari matritsani qulay usulda toping:

$$a)\begin{pmatrix} 1 & 5 & 7 \\ 3 & 1 & 1 \\ 2 & 3 & 4 \end{pmatrix};\quad b)\begin{pmatrix} 2 & -1 & 7 \\ 5 & 3 & 2 \\ 1 & 4 & 3 \end{pmatrix};\quad c)\begin{pmatrix} 1 & 0 & -2 \\ 3 & 1 & 0 \\ -1 & 2 & 4 \end{pmatrix};\quad e)\begin{pmatrix} 3 & 2 & 2 \\ 1 & 3 & 1 \\ 5 & 3 & 4 \end{pmatrix}$$

$$d)\begin{pmatrix} 1 & 1 & 1 & \dots & 1 \\ 0 & 1 & 1 & \dots & 1 \\ 0 & 0 & 1 & \dots & 1 \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & 1 \end{pmatrix};\quad f)\begin{pmatrix} 1 & 0 & 0 & 0 & \dots & 0 & 0 \\ a & 1 & 0 & 0 & \dots & 0 & 0 \\ 0 & a & 1 & 0 & \dots & 0 & 0 \\ 0 & 0 & a & 1 & \dots & 0 & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & 0 & \dots & a & 1 \end{pmatrix};$$

$$g)\begin{pmatrix} 1 & a & a^2 & a^3 & \dots & a^n \\ 0 & 1 & a & a^2 & \dots & a^{n-1} \\ 0 & 0 & 1 & a & \dots & a^{n-2} \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & 0 & \dots & 1 \end{pmatrix}$$

3.5. Matritsa rangini ta'rifga asosan hisoblang: $A=\begin{pmatrix} 2 & -1 & 3 & -2 & 4 \\ 4 & -2 & 5 & 1 & 7 \\ 2 & -1 & 1 & 8 & 2 \end{pmatrix}$.

Yechish. A matritsa $3-$tartibli, demak uning rangi 3 dan yuqori bo'lmaydi. Uchinchi tartibli minorlarni hisoblaymiz:

$$M_1=\begin{vmatrix} 2 & -1 & 3 \\ 4 & -2 & 5 \\ 2 & -1 & 1 \end{vmatrix}=-4-10-12+12+4+10=0$$

$$M_2=\begin{vmatrix} 2 & -1 & -2 \\ 4 & -2 & 1 \\ 2 & -1 & 8 \end{vmatrix}=-32-2+8-8+32+2=0$$

$$M_3=\begin{vmatrix} 2 & -1 & 4 \\ 4 & -2 & 7 \\ 2 & -1 & 2 \end{vmatrix}=-8-14-16+16+8-14=0$$

$$M_4 = \begin{vmatrix} -1 & 3 & -2 \\ -2 & 5 & 1 \\ -1 & 1 & 8 \end{vmatrix} = -40 - 3 + 4 - 10 + 48 + 1 = 0$$

$$M_5 = \begin{vmatrix} 3 & -2 & 4 \\ 5 & 1 & 7 \\ 1 & 8 & 2 \end{vmatrix} = 6 - 14 + 160 - 4 + 20 - 168 = 0$$

Barcha uchinchi tartibli minorlar nolga teng. Ikkinchi tartibli minorlarni hisoblaymiz:

$$M_1^1 = \begin{vmatrix} -1 & 3 \\ -2 & 5 \end{vmatrix} = -5 + 6 = 1 \qquad M_1^1 \neq 0, \qquad r(A) = 2.$$

Bu usulda noldan farqli minor topilgunga qadar hisoblashlar davom etadi. Shuning uchun $3-$tartibli kattaroq matritsa rangini hisoblash bir muncha qiyinchiliklarga olib keladi. ▶

3.6. Matritsa rangini elementar almashtirishlar yordamida nollar yig‘ib hisoblang:

$$A = \begin{pmatrix} 25 & 31 & 17 & 43 \\ 75 & 94 & 53 & 132 \\ 75 & 94 & 54 & 134 \\ 25 & 32 & 20 & 48 \end{pmatrix}$$

Yechish. $$A = \begin{pmatrix} 25 & 31 & 17 & 43 \\ 75 & 94 & 53 & 132 \\ 75 & 94 & 54 & 134 \\ 25 & 32 & 20 & 48 \end{pmatrix} \sim \begin{pmatrix} 25 & 31 & 17 & 43 \\ 0 & 1 & 2 & 3 \\ 0 & 1 & 3 & 5 \\ 0 & 1 & 3 & 5 \end{pmatrix} \sim \begin{pmatrix} 25 & 31 & 17 & 43 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Bu matritsaning rangi $\begin{pmatrix} 25 & 31 & 17 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}$ matritsa rangiga teng.

$$\begin{vmatrix} 25 & 31 & 17 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{vmatrix} = 25 \neq 0 \qquad r\begin{pmatrix} 25 & 31 & 17 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix} = 3$$

Demak, berilgan matritsaning rangi ham 3 ga teng. $r(A)=3$. ▶

? Mashqlarni bajaring.

3.7. Berilgan kvadrat matritsaning rangini toping. Xosmas matritsaning teskarisini toping:

$$a)\ A=\begin{pmatrix}1 & 2\\ -2 & 0\end{pmatrix};\qquad b)\ B=\begin{pmatrix}2 & 5\\ -4 & 2\end{pmatrix};$$

$$c)\ C=\begin{pmatrix}-1 & 0 & 8\\ 5 & 9 & 0\\ 0 & 4 & 3\end{pmatrix};\qquad d)\ D=\begin{pmatrix}2 & 1 & 2\\ 1 & 1 & 1\\ 2 & 3 & 2\end{pmatrix};$$

$$f)\ F=\begin{pmatrix}2 & 3 & 4 & 0\\ 1 & 5 & 7 & 0\\ 3 & 1 & 1 & 0\\ 0 & 0 & 0 & 1\end{pmatrix};\qquad g)\ G=\begin{pmatrix}1 & 2 & 1 & 0\\ 1 & 1 & 3 & 1\\ 1 & 2 & 1 & 1\\ 1 & 1 & 3 & 0\end{pmatrix}$$

3.8. Quyidagi matritsalar rangini minorlar ajratish usuli bilan hisoblang:

$$a)\begin{pmatrix}3 & 5 & 7\\ 1 & 2 & 3\\ 1 & 3 & 5\end{pmatrix};b)\begin{pmatrix}1 & 2 & 3 & 6\\ 2 & 3 & 1 & 6\\ 3 & 1 & 2 & 6\end{pmatrix};c)\begin{pmatrix}1 & 2 & 3 & 4\\ 2 & 4 & 6 & 8\\ 3 & 6 & 9 & 12\end{pmatrix};d)\begin{pmatrix}0 & 2 & 0 & 0\\ 1 & 0 & 0 & 4\\ 0 & 0 & 3 & 0\end{pmatrix}$$

$$e)\begin{pmatrix}2 & -4 & 3 & 1 & 0\\ 1 & -2 & 1 & -4 & 2\\ 0 & 1 & -1 & 3 & 1\\ 4 & -7 & 4 & -4 & 5\end{pmatrix};j)\begin{pmatrix}1 & 2 & 1 & 3\\ 4 & -1 & -5 & -6\\ 1 & -3 & -4 & -7\\ 2 & 1 & -1 & 0\end{pmatrix};k)\begin{pmatrix}0 & 2 & -4\\ -1 & -4 & 5\\ 3 & 1 & 7\\ 0 & 5 & -10\\ 2 & 3 & 0\end{pmatrix}$$

? Quyidagi 3.9 – 3.19 – misollarda matritsalar rangini elementar almashtirish usuli bilan hisoblang:

3.9. $\begin{pmatrix} 1 & 2 & 1 & 3 & 4 \\ 3 & 4 & 2 & 6 & 8 \\ 1 & 2 & 1 & 8 & 4 \end{pmatrix}.$

3.10. $\begin{pmatrix} 1 & 7 & 5 & 8 & 9 & 2 \\ 3 & 21 & 15 & 24 & 27 & 6 \\ 2 & 14 & 10 & 16 & 18 & 4 \end{pmatrix}.$

3.11. $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 9 & 12 \end{pmatrix}.$

3.12. $\begin{pmatrix} 1 & 0 & 2 & 0 & 0 \\ 0 & 1 & 0 & 2 & 0 \\ 2 & 0 & 4 & 0 & 0 \end{pmatrix}.$

3.13. $\begin{pmatrix} 4 & 3 & -5 & 2 & 3 \\ 8 & 6 & -7 & 4 & 2 \\ 4 & 3 & -8 & 2 & 7 \\ 4 & 3 & 1 & 2 & -5 \\ 8 & 6 & -1 & 4 & -6 \end{pmatrix}.$

3.14. $\begin{pmatrix} 24 & 19 & 36 & 72 & -38 \\ 49 & 40 & 73 & 147 & -80 \\ 73 & 59 & 98 & 219 & -118 \\ 47 & 36 & 71 & 141 & -72 \end{pmatrix}.$

3.15. $\begin{pmatrix} 17 & -28 & 45 & 11 & 39 \\ 24 & -37 & 61 & 13 & 50 \\ 25 & -7 & 32 & -18 & -11 \\ 31 & 12 & 19 & -43 & -55 \\ 42 & 13 & 29 & -55 & -68 \end{pmatrix}.$

3.16. $\begin{pmatrix} 47 & -67 & 35 & 201 & 155 \\ 26 & 98 & 23 & -294 & 6 \\ 16 & -428 & 1 & 1284 & 52 \end{pmatrix}.$